



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi & Affiliated to Anna University

Accredited by NAAC & An ISO 9001: 2015 Certified Institution

NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year 2023 – 2024 (Even Semester)

Degree, Semester & Branch : IV Semester B.Tech-IT
Course Code & Title : CS3451 & Introduction to Operating Systems
Name of the Faculty member : Mr. Balaganesh M V, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit V / Virtualization and Operating System Components
- **Course Outcome:** CO5
- **Topic Learning Outcome:** TLO 14
- **Activity Chosen:** Collaborative Learning
- **Justification:**

The term virtualization has many meanings, and aspects of virtualization permeate all aspects of computing. Virtual machines are one instance of this trend. Virtual machine implementations involve several components. The prime objective of Collaborative Learning is to promote both self and peer teaching which requires students to understand the concept very clearly and engage in discussion and learning.

- **Time Allotted for the Activity:** 20 Minutes

- **Details of the Implementation:**

The course instructor initially chose five student volunteers depending on their class involvement, interest, and subject knowledge. The leaders are allowed to make groups on their own. A Virtualization and Operating System Components topic has been assigned to the volunteers on the spot of the activity. On the day of the activity, the students are seated with their group members and an individual problem statement has been given to each team leader. The leaders have solved the problem on their own and discuss with their team members about the solution. The team members are also actively involved in the discussion and clarify their doubts with their leader. Finally, the instructor consolidated the information that was discussed in this activity for another 10 minutes as shown in Figure 1 & 2.

- **CO – PO / PSO mapping specific to this event:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PSO2
CO5	3	3	2	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

• **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO4	PO9	PO10	PSO2
	3	3	2	1	1	1	1
Justification for correlation	Apply the knowledge of virtualization to solve problems	To analyze complex virtualization problems	To design a Virtual machine monitor.	Able to choose appropriate virtual machine configuration for the given resource	To work as an individual, and as a member	To Communicate effectively on complex engineering activities	A basic understanding of virtualization will help the students to understand cloud technologies

• **Images / Screenshot of the practice:**



Fig 1. Karthik Raja E sharing the solution with Sri Saru Kumar S, Prasath J, Sridhar S, Praveen Kumar R and Harikrishnan K.



Fig 2. Ashmitha K V sharing the solution to Harshini G, Elakiya E and Suba Lakshmi G

• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- Some of the students were felt the activity is very interesting.
- The activity conducted gave them a chance to improve their communication skills, also working as a team they can face group discussions in future workplaces.
- Through learning from peer members, students felt comfortable clarifying their doubts.

❖ **Benefit of the practice:**

- Through this activity the students were made to work as a team and explain to others also, like a group discussion activity.
- The students are made to understand the different applications through knowledge sharing.

❖ ***Challenges faced in implementation:***

- Some students in the team did not participate actively in the team.
- All the students were not able to understand the memory management concept in virtualization. some students felt it was very difficult to do this activity. After the activity, I explained to the particular students in the class.

References:

- ❖ <https://teaching.cornell.edu/teaching-resources/engaging-students/collaborative-learning>

Signature of Faculty Member

HOD